NTIRE 2026 CD-FSOD Challenge Factsheet
Negative Prompting for Few-Shot Object Detection

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1. Factsheet Information

1.1. Team details

- Team name: J_G_team
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- Team members: Guangwei Huang
- Method name: Negative Prompting for Few-Shot Object Detection
- Final score: 149.95
- Submission link: https://www.codabench.org/competitions/12873

1.2. Method overview

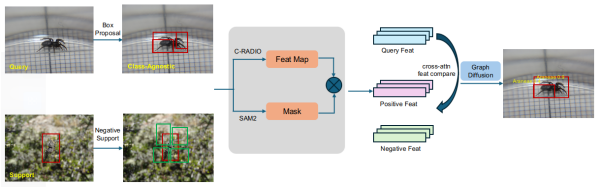

Figure 1. Overall pipeline of Negative Prompting for Few-Shot Object Detection.

High-level idea: We propose a few-shot object detection method based on FSOD-VFM [1] with two key improvements. First, we incorporate negative sample features generated by randomly shifting and scaling positive bounding boxes, providing the model with additional background context. Second, we replace static class prototypes with query-adaptive cross-attention prototypes, allowing support features to be dynamically weighted according to each query proposal. These two strategies improve the model's ability to discriminate between foreground and background, and enable it to adapt to intra-class variation, leading to more accurate detection.

Key contributions:

- **Contribution 1:** We introduce negative sample features as additional guidance, helping the model distinguish foreground from background.
- **Contribution 2:** We propose cross-attention prototypes that dynamically adapt to each query proposal, improving matching precision.

1.3. Model architecture

- **Detector:** Universal Proposal Network [2] is used to generate category-agnostic bounding box proposals for candidate objects.
- **Mask extraction module:** SAM2 is employed to produce accurate object masks for support annotations, aiding robust feature extraction.
- **Visual Feature Extractor:** We use C-RADIOv4-H as visual feature extractor for whole image.

1.4. Training details

This method is training-free and does not require any task-specific finetuning.

1.5. Inference details

- Test-time settings: During preprocessing, C-RADIOv4 resizes all input images so that the longer side is 1260. For all proposals generated by UPN, we select top 100 boxes per image ranked by confidence.

1.6. Experimental results

Experimental results is shown in Table.1.

2. Other details

- Planned submission of a solution description paper at NTIRE 2026 workshop: No.

	Method	Setting	Dataset1	Dataset2	Dataset3
1-shot	FSOD-VFM	(Open)	26.93	58.84	17.28
	Ours	(Open)	26.71	57.99	18.01
5-shot	FSOD-VFM	(Open)	38.21	58.70	25.80
	Ours	(Open)	38.46	57.94	26.77
10-shot	FSOD-VFM	(Open)	35.03	58.21	27.31
	Ours	(Open)	34.85	57.50	28.86

Table 1. Results under 1/5/10-shot settings on three target datasets. Replace placeholders with official numbers.

- General comments and impressions of the NTIRE 2026 challenge: This challenge provides a valuable benchmark for evaluating few-shot object detection under cross-domain scenarios. The task setup is meaningful and reflects practical challenges in real-world applications.
- Suggestions for future CD-FSOD challenges: Future editions of the challenge could explore more diverse domain shifts or introduce additional subtracks focusing on efficiency.

References

[1] Chen-Bin Feng, Youyang Sha, Longfei Liu, Yongjun Yu, Chi Man Vong, Xuanlong Yu, and Xi Shen. Few-shot object detection with vision foundation models and graph diffusion. In *The Fourteenth International Conference on Learning Representations*, 2026. 1

[2] Qing Jiang, Gen luo, Yuqin Yang, Yuda Xiong, Yihao Chen, Zhaoyang Zeng, Tianhe Ren, and Lei Zhang. Chatrex: Taming multimodal llm for joint perception and understanding, 2024. 1